

A Multi-Modal Off-Axis Polarized Digital Holography Microscopy System for Depth-Resolved Stress and 3D Profile Metrology for Advanced Packaging

Project Overview

This project focuses on the development of a hybrid optical metrology platform for the inspection of Through-Glass Via (TGV) structures used in advanced semiconductor packaging. The system combines off-axis digital holographic microscopy and polarization-based photoelasticity to measure both the 3D structural profile and stress distribution of transparent glass samples.

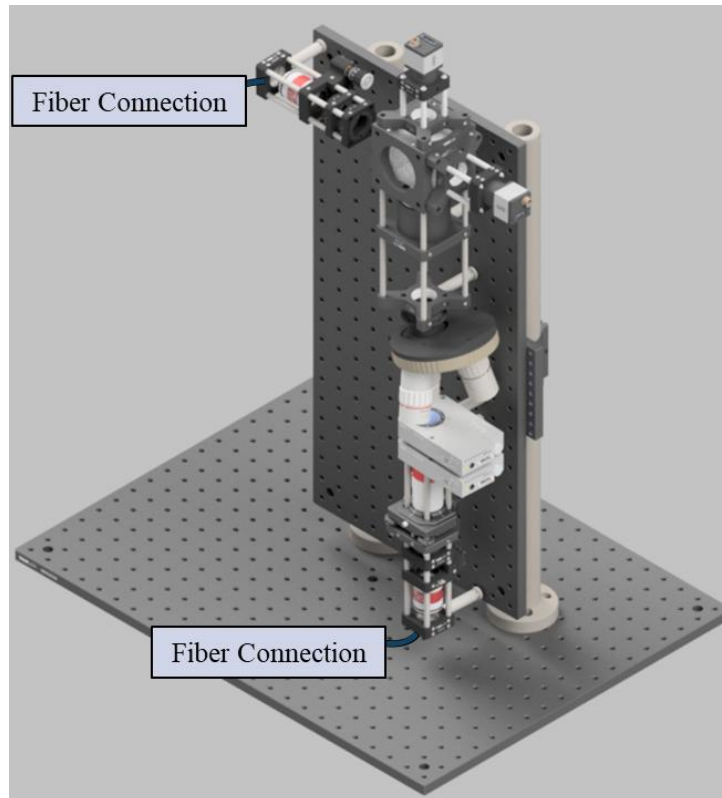


Figure 1. CAD Design of the experimental probe system.

Conventional inspection techniques often require separate systems for dimensional metrology and stress analysis. In contrast, this project aims to integrate both capabilities into a single optical platform, enabling non-destructive characterization of transparent samples with potential applications in semiconductor packaging, glass substrate inspection, and industrial quality control.

Motivation

Through-Glass Vias are becoming increasingly important in advanced packaging because they enable vertical interconnects through glass substrates. However, laser modification, wet etching, and subsequent processing steps can introduce residual stress, micro-defects, and geometrical deviations. These issues may affect mechanical reliability, yield, and long-term device performance.

The main motivation of this project was to develop an optical inspection approach capable of evaluating:

- 3D profile and surface/topographic information of TGV's
- Residual stress distribution of laser modified samples and TGV's
- Defect detection and reconstruction
- Potential non-destructive inline metrology applications.

System Concept

The proposed system combines two complementary optical techniques:

Off-axis digital holographic microscopy is used to reconstruct the complex optical field of the sample. From the reconstructed phase and amplitude, the system can extract structural information and perform numerical refocusing for depth-resolved analysis.

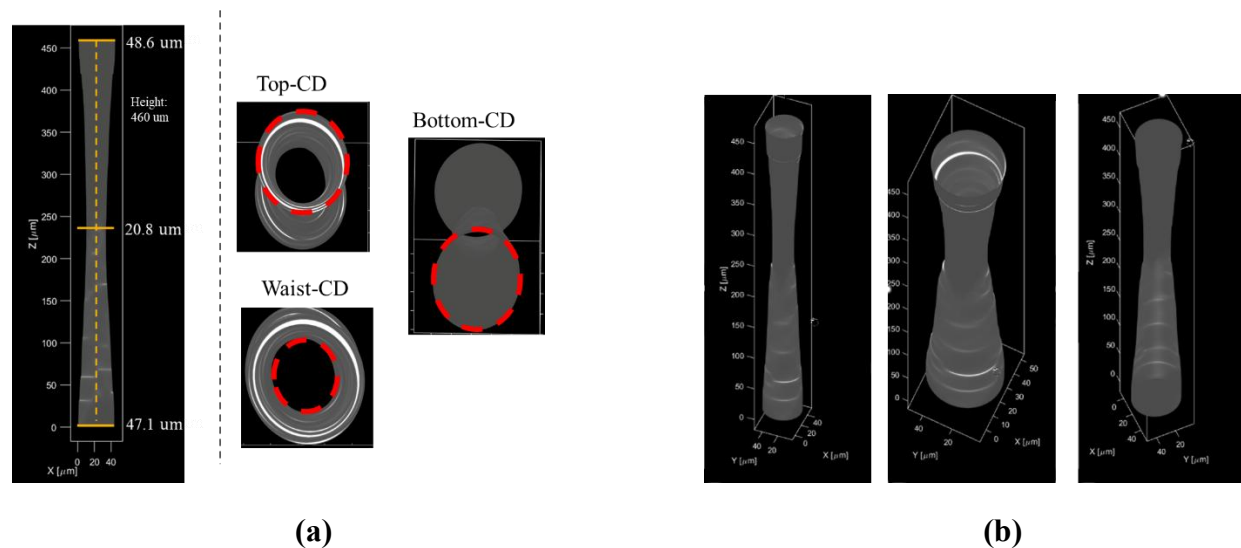


Figure 2. (a) TGV Critical Dimensions and (b) 3D profile reconstruction. Results obtained using the proposed system.

Polarization imaging and digital photoelasticity are used to evaluate stress-induced birefringence in transparent materials. By analyzing polarization-resolved intensity information, the system estimates optical retardation and converts it into stress-related information.

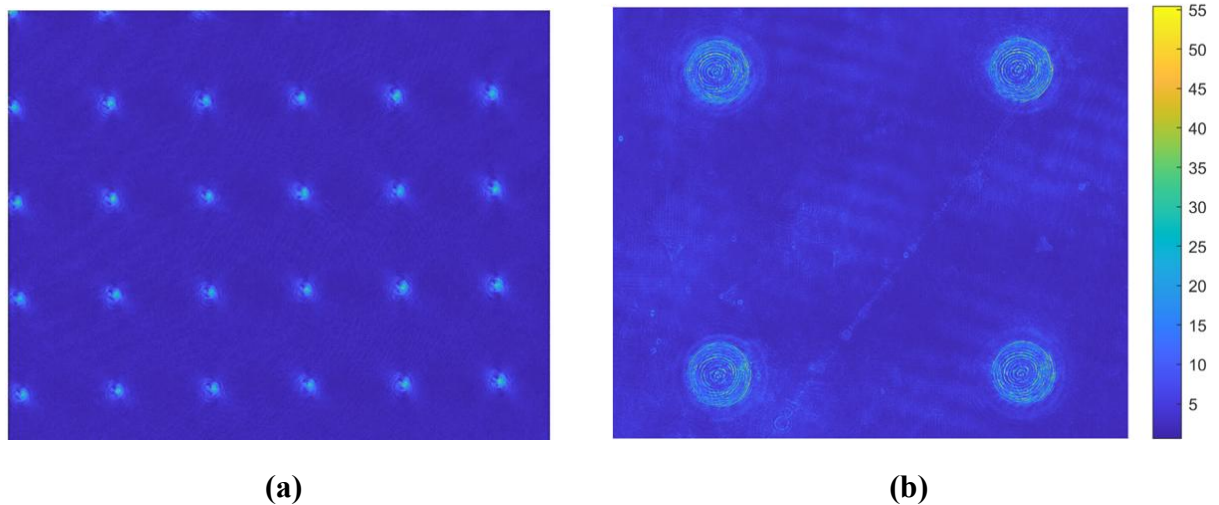


Figure 3. (a) Stress measurement of laser-modified sample (b) Stress measurement of TGV (after wet etching process). Results obtained using the proposed system.

By integrating these two approaches, the system provides a multi-modal workflow for both structural and stress characterization.

My Role

My contributions to this project included:

- Designed and implemented the experimental optical system.
- Selected optical and optomechanical components for the setup.
- Prepared CAD models and system layouts for system integration.
- Integrated polarization imaging, digital photoelasticity, and digital holography into a unified metrology workflow.
- Developed MATLAB-based image processing routines for phase reconstruction, polarization analysis, retardation-to-stress mapping, and visualization.
- Performed optical measurements on TGV industrial samples.
- Prepared technical presentations to communicate the system design, measurement capabilities, and experimental results to industrial partners.

Technical Workflow

The measurement workflow can be summarized as follows:

1. Illuminate the transparent sample using a controlled optical source.
2. Acquire holographic and polarization-resolved image data.
3. Reconstruct the complex optical field from the digital hologram.
4. Extract amplitude and phase information from the reconstructed optical field.
5. Analyze polarization-resolved signals to estimate retardation.
6. Convert retardation information into stress-related maps using photoelastic principles.
7. Perform numerical refocusing or depth-dependent analysis for 3D characterization.
8. Visualize stress distribution, phase information, and reconstructed sample profiles.

Key Technical Areas

This project demonstrates experience in:

- Digital holographic microscopy
- Polarization imaging
- Digital photoelasticity
- Optical stress measurement
- Phase reconstruction
- Retardation-to-stress mapping
- Numerical refocusing
- Optical system alignment
- MATLAB-based image processing
- Advanced packaging metrology
- Through-Glass Via inspection

Results and Demonstrations

The system was tested using transparent glass samples and TGV-related structures. Experimental results demonstrated the ability to visualize stress-related patterns and reconstruct optical phase information from measured samples.

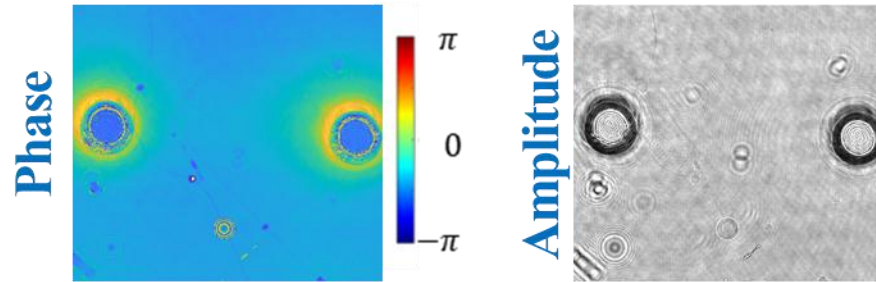


Figure 4. Reconstructed phase or amplitude image. Results obtained using the proposed system.

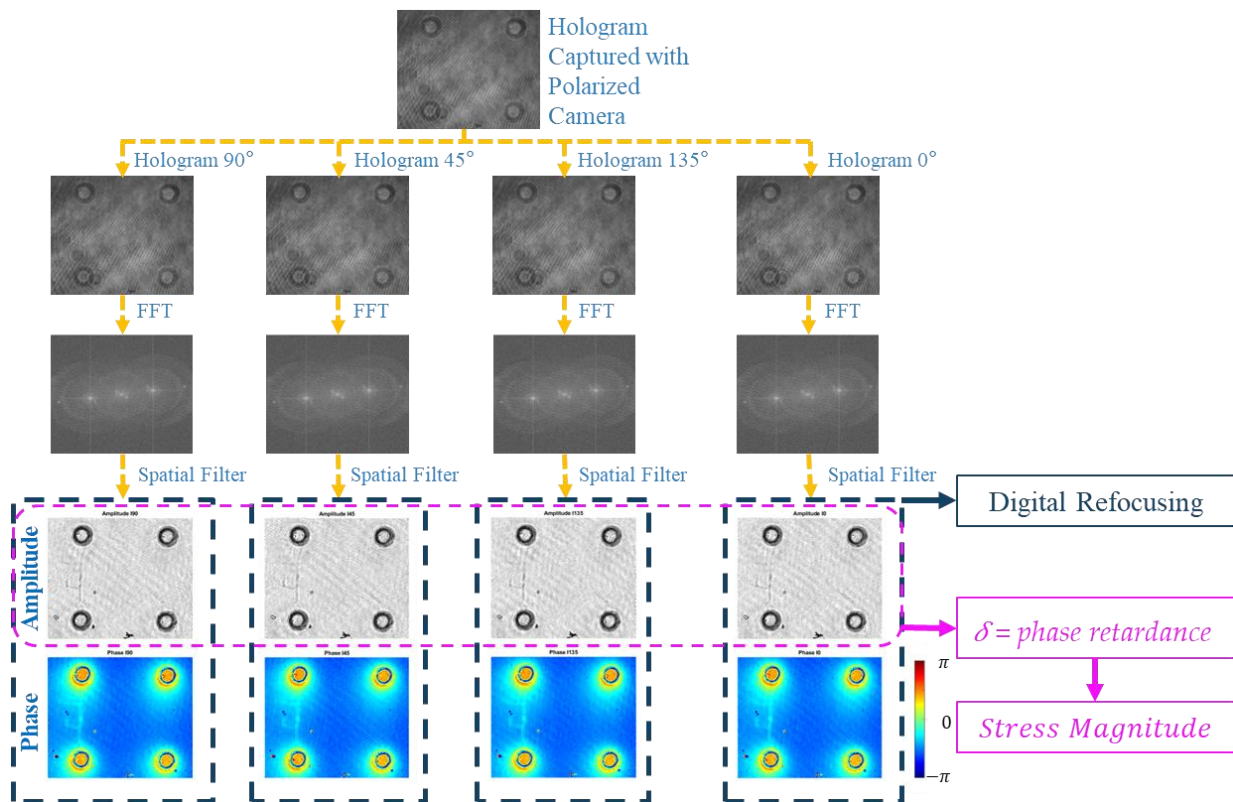


Figure 5. Processing workflow from image acquisition to stress/profile reconstruction using DHM system. Results obtained using the proposed system.

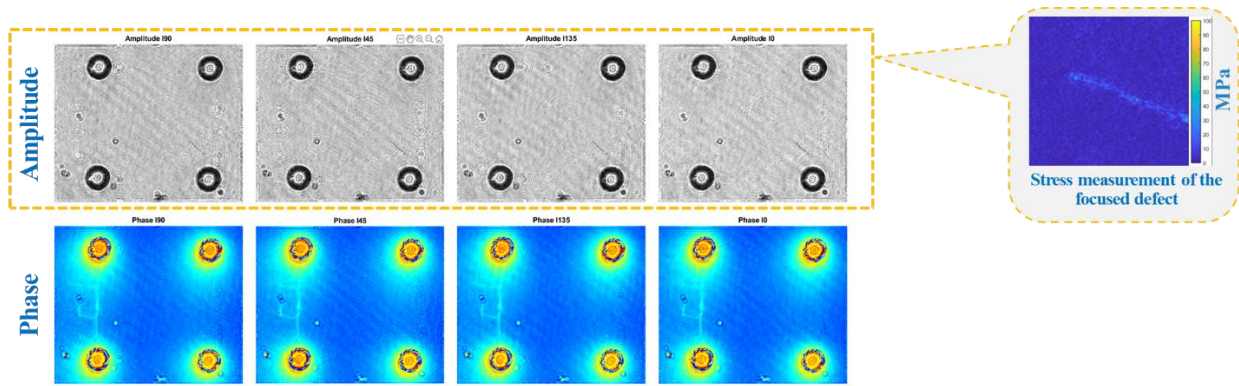


Figure 6. Depth resolved stress measurement from stress reconstruction using DHM system after digital refocusing using angular spectrum method. Results obtained using the proposed system.

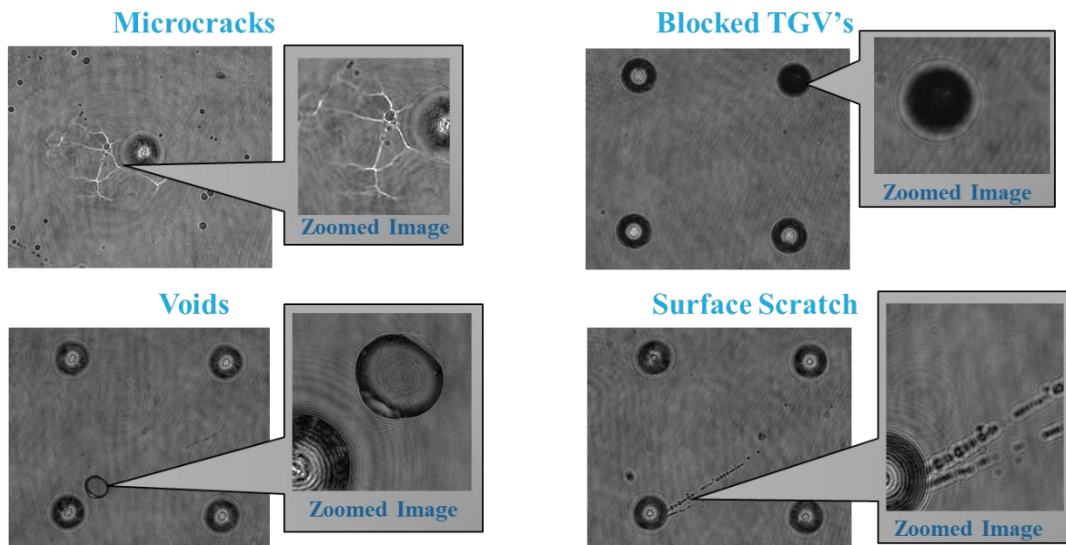


Figure 7. Defect detection. Results obtained using the proposed system.

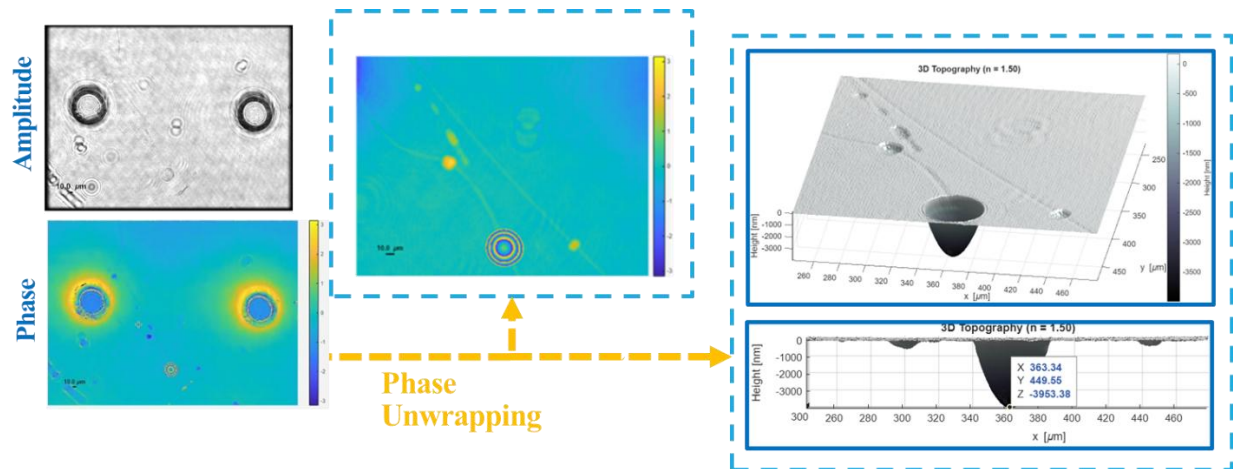


Figure 8. Defect reconstruction from phase unwrapping using DHM system. Results obtained using the proposed system.

Industrial Relevance

The system was developed with semiconductor and transparent material inspection in mind. It supports non-destructive optical characterization and can be applied to samples relevant to advanced packaging, display technology, and precision glass inspection.

Compared with conventional inspection workflows that often separate stress analysis and dimensional metrology, this project explores a combined optical approach that can provide complementary information from a single metrology platform.

Confidentiality Note

Some technical details, raw experimental data, sample information, and partner names are omitted or generalized due to confidentiality and intellectual property considerations.

Project Status

- Graduate thesis project
- Patent application filed by National Taiwan University
- Related conference publications and presentations
- Ongoing development for advanced optical metrology applications